2024 JUNE PAPER 1
M5 BY GRANDMASTER
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(1) $U_{n+1} = 3U_n + 1$

First 4 terms starting with U_2 given that $U_1 = 0$

$U_{1+1} = 3U_1 + 1$
$U_2 = 3(0) + 1$
$U_2 = 1$

$U_3 = 3U_2 + 1$
$U_3 = 3(1) + 1$
$U_3 = 4$

$U_4 = 3U_3 + 1$
$U_4 = 3(4) + 1$
$U_4 = 13$

$U_5 = 3(U_4) + 1$
$U_5 = 3(13) + 1$
$U_5 = 40$

$\therefore$ First 4 terms are $\{1, 4, 13, 40\}$

(ii) It is diverging

(iii) U_3 in terms of U_1
$U_3 = 3U_2 + 1$
But as you can see, that we need to express U_3 in terms of U_1

So $U_2 = 3U_1 + 1$

Now $U_3 = 3(3U_1 + 1) + 1$
$U_3 = 9U_1 + 4$

(2) $y = ae^{bx}$
$\ln y = \ln(ae^{bx})$
$\ln y = \ln a + \ln e^{bx}$
$\ln y = \ln a + bx$
$\ln y = bx + \ln a$
$y = mx + c$

$m = b$

[where m is the gradient]

Now finding m

$P_1 = (0, \ln 2)$ $P_2 = (3, \ln 5)$

$m = \dfrac{\ln 5 - \ln 2}{3}$

$m = \dfrac{1}{3}\ln\left(\dfrac{5}{2}\right)$

But $m = b$

$\therefore b = \dfrac{1}{3}\ln\left(\dfrac{5}{2}\right)$

$\ln a = c$

where c is the y intercept

So $\ln a = \ln 2$

exponent both sides

$a = 2$

(3) $\dfrac{2x-3}{x^2(x^2-4)} = \dfrac{2x-3}{x^2(x+2)(x-2)}$

To Partial Fraction

$$\frac{2x-3}{x^2(x+2)(x-2)} = \frac{Ax+B}{x^2} + \frac{C}{x+2} + \frac{D}{x-2}$$

$2x-3 = (Ax+B)(x+2)(x-2) + C(x^2)(x-2) + D(x^2)(x+2)$

when $x=-2$

$2(-2)-3 = (A(-2)+B)(2)(0) + C(-2)^2(-4)$

$-4-3 = C(4)(-4)$

$-7 = -16C$

$C = \dfrac{7}{16}$

when $x = 2$

$2(2)-3 = D(2)^2(4)$

$1 = 16D$

$D = \dfrac{1}{16}$

when $x=0$

$-3 = B(2)(-2)$

$-3 = -4B$

$B = \dfrac{3}{4}$

when $x = 1$

$2-3 = (A+B)(3)(-1) + C(1)(-1) + D(1)(3)$

$-1 = -3A - 3B - C + 3D$

$3A = -3\left(\frac{3}{4}\right) - \frac{7}{16} + 3\left(\frac{1}{16}\right) + 1$

$A = \dfrac{-1}{2}$

$$\therefore \frac{2x-3}{x^2(x^2-4)} = \frac{-2x+3}{4x^2} + \frac{7}{16(x+2)} + \frac{1}{16(x-2)}$$

(4) $\dfrac{e^{2x}+e^{-2x}}{5} = 1$

$e^{2x}+e^{-2x} = 5$

$e^{2x} + \dfrac{1}{e^{2x}} = 5$

$e^{4x} - 5e^{2x} + 1 = 0$

$$e^{2x} = \frac{5 \pm \sqrt{(-5)^2 - 4(1)(1)}}{2(1)}$$

$$e^{2x} = \frac{5 \pm \sqrt{21}}{2}$$

natural log both sides

$2x = \ln\left(\dfrac{5 \pm \sqrt{21}}{2}\right)$

$x = \dfrac{1}{2}\ln\left(\dfrac{5 \pm \sqrt{21}}{2}\right)$

$x = \ln\sqrt{\dfrac{5 \pm \sqrt{21}}{2}}$

$x = \ln\left(\dfrac{\sqrt{7} \pm \sqrt{3}}{2}\right)$

(5) $(x-3)^2 > 2x+1$

$x^2-6x+9-2x-1>0$

$x^2-8x+8>0$

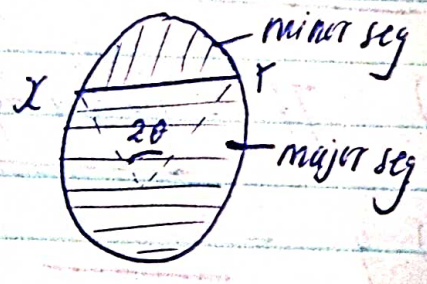
$$x = \frac{8 \pm \sqrt{(-8)^2-4(1)(8)}}{2(1)}$$

$$x = 4 \pm 2\sqrt{2}$$

$x < 4-2\sqrt{2}$	$4-2\sqrt{2} < x < 4+2\sqrt{2}$	$x > 4+2\sqrt{2}$
+	−	+

$\therefore$ $x < 4-2\sqrt{2} \cup x > 4+2\sqrt{2}$

(6)



Minor segment $= \frac{1}{6}$ major segment

$$\frac{1}{2}r^2(2\theta-\sin 2\theta) = \frac{1}{6}\left[\pi r^2 - \frac{1}{2}r^2(2\theta-\sin 2\theta)\right]$$

$$3r^2(2\theta-\sin 2\theta) = \pi r^2 - \frac{1}{2}r^2(2\theta-\sin 2\theta)$$

$$\frac{7}{2}r^2(2\theta-\sin 2\theta) = \pi r^2$$

$$\frac{7}{2}\cdot 2\theta - \frac{7}{2}\sin 2\theta = \pi$$

$$7\theta = \pi + \frac{7}{2}\sin 2\theta$$

$$\frac{7}{2}\cdot\sin 2\theta = 7\theta - \pi$$

$$\sin 2\theta = \frac{2}{7}(7\theta - \pi)$$

as required

(7a) $$(1-2x)^{1/2} = 1 + (1/2)(-2x) + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)(-2x)^2}{2!}$$

$$\therefore (1-2x)^{1/2} = 1 - x - \frac{x^2}{2}$$

(b) $|-2x| < 1$

$-1 < -2x < 1$

$$-\frac{1}{2} < x < \frac{1}{2}$$

(b) putting $x = 1/9$

$$\left(1-2\left(\frac{1}{9}\right)\right)^{1/2} = \text{RHS}$$

$$\frac{\sqrt{7}}{3} = \text{RHS}$$

$$\sqrt{7} \approx 3\left[1-\left(\frac{1}{9}\right)-\frac{(1/9)^2}{2}\right]$$

$$\sqrt{7} \approx 3\left(\frac{143}{162}\right)$$

$$\sqrt{7} \approx \frac{143}{54}$$ as required

(8) $l \Rightarrow y + 3x - 9 = 0$

$y = -3x + 9$

gradient of $l = -3$

perpendicular lines

$m_1 \times m_2 = -1$

$-3m = -1$

$m = \frac{1}{3}$

gradient of line $m = \frac{1}{3}$

Point A $(1;6)$

$y - 6 = \frac{1}{3}(x - 1)$

$3y - 18 = x - 1$

$3y = x + 17$

line $m \Rightarrow$ $x - 3y + 17 = 0$

(b) parallel lines have same gradient.

point B $(5;6)$

$y - 6 = -3(x - 5)$

$y - 6 = -3x + 15$

$y = -3x + 21$

(c) $p = \left| \frac{x - 3y + 17}{\sqrt{(1)^2 + (-3)^2}} \right|$

point B $(5;6)$

$p = \left| \frac{5 - 3(6) + 17}{\sqrt{1 + 9}} \right|$

$p = \frac{2\sqrt{10}}{5}$ units

(9) $x^2 + 6x + 5 \equiv (ax + b)^2 + c$

$(x+3)^2 - 9 + 5 = RHS$

$(x+3)^2 - 4 = RHS$

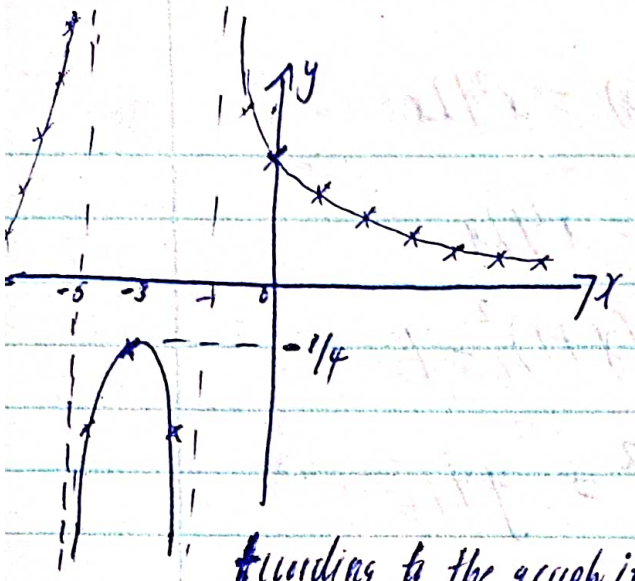
(b) TP $= (-3; -4)$

(c) $y = \frac{1}{x^2 + 6x + 5}$

$y = \frac{1}{(x+3)^2 - 4}$

We can't sketch an appropriate graph using transformations so we need to use old school.

-5	-4	-3	-2	-1	0	1	2	3	4
ERRO	$-1/3$	$-1/4$	$-1/3$	ere	$1/5$	$1/12$	$1/21$	$1/32$	$1/45$



According to the graph it shows us the range of values are

$$\boxed{\begin{array}{l} y > 0 \\ \text{for } x > -1 \text{ and } x < -5 \end{array}}$$

$$\boxed{\begin{array}{l} y \leq -1/4 \\ \text{for } -5 < x < -1 \end{array}}$$

Some may want to take this shortcut.

Finding the Inverse

$$\frac{1}{y} = (x+3)^2 - 4$$

$$x+3 = \sqrt{\frac{1}{y} + 4}$$

$$x = -3 + \sqrt{\frac{1}{y} + 4}$$

$$f^{-1}(x) = -3 + \sqrt{\frac{1}{x} + 4}$$

looking for the domain

$$\frac{1}{x} + 4 \geq 0$$

$$\frac{1}{x} \geq -4$$

$$x \geq -4x^2$$

$$4x^2 + x \geq 0$$

$$x(4x+1) \geq 0$$

C.V

$0 \quad ; \quad -1/4$

$x \leq -1/4$	$-1/4 \leq x \leq 0$	$x \geq 0$
+	−	+

$$x \leq -1/4 \cup x \geq 0$$

Domain $f^{-1}(x)$ = Range $f(x)$

$\therefore$ Range $f(x)$ =

$$y \leq -\frac{1}{4} \text{ and } y \geq 0$$

Please note that $y \neq 0$

So we reject this method

(10) $f(x) = x^4 + x^3 + 23x^2 + 25x - 50$

$f(-5i) = 0$

$$(-5i)^4 + (-5i)^3 + 23(-5i)^2 + 25(-5i) - 50 = 0$$

$$625i^4 - 125i^3 + 575i^2 - 125i - 50 = 0$$

$$625 + 125i - 575 - 125i - 50 = 0$$

$$0 = 0$$

$\therefore -5i$ is a root shown

(b) $x^2-(0)x+25=0$

$$
\begin{array}{r|l}
 & x^2+x-2 \\
\hline
x^2+25 & x^4+x^3+23x^2+25x-50 \\
 & -(x^4 \quad +25x^2) \\
 & x^3-2x^2+25x \\
 & -(x^3 \quad +25x) \\
 & -2x^2 \quad -50 \\
 & -(-2x^2 \quad -50) \\
 & 0
\end{array}
$$

$(x^2+25)(x^2+x-2)=0$

$(x^2+25)(x+2)(x-1)=0$

$x = 5i, -5i, 1$ and -2

(11a) $gh(x) = g\left(\frac{1}{x-3}\right)$

$4\left(\frac{1}{x-3}\right)+1$

$\frac{4}{x-3}+1$

$\frac{4+x-3}{x-3}$

$\frac{1+x}{x-3}$; domain $x \neq 3$

(ii) $f(x) = x^2+2x$

$y = x^2+2x$

$y = (x+1)^2-1$

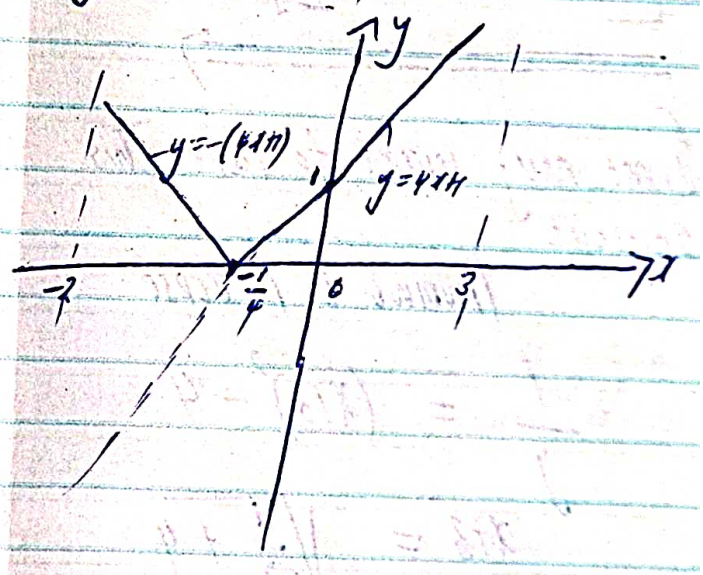
$(x+1)^2 = y+1$

$x = -1+\sqrt{y+1}$

$f^{-1}(x) = -1+\sqrt{x+1}$

(bi) $y = |g(x)|$

$y = |4x+1|$



(ii) $|g(x)| \geq 0$

(12i) $\frac{\cos\theta}{1-\cos\theta} - \frac{\cos\theta}{1+\cos\theta} \equiv 2\cot^2\theta$

LHS

$$\frac{\cos\theta(1+\cos\theta) - \cos\theta(1-\cos\theta)}{(1-\cos\theta)(1+\cos\theta)}$$

$$\frac{\cos\theta + \cos^2\theta - \cos\theta + \cos^2\theta}{1-\cos^2\theta}$$

$$\frac{2\cos^2\theta}{\sin^2\theta + \cos^2\theta - \cos^2\theta}$$

$$\frac{2\cos^2\theta}{\sin^2\theta}$$

$2\cot^2\theta$ = RHS proved

(ii)

$2\cot^2\theta = 1$

$\cot\theta = \pm\sqrt{\frac{1}{2}}$

$\frac{1}{\tan\theta} = \pm\sqrt{\frac{1}{2}}$

$\tan\theta = \pm\frac{2}{\sqrt{2}}$

$PV = \tan^{-1}\left(\pm\frac{2}{\sqrt{2}}\right)$

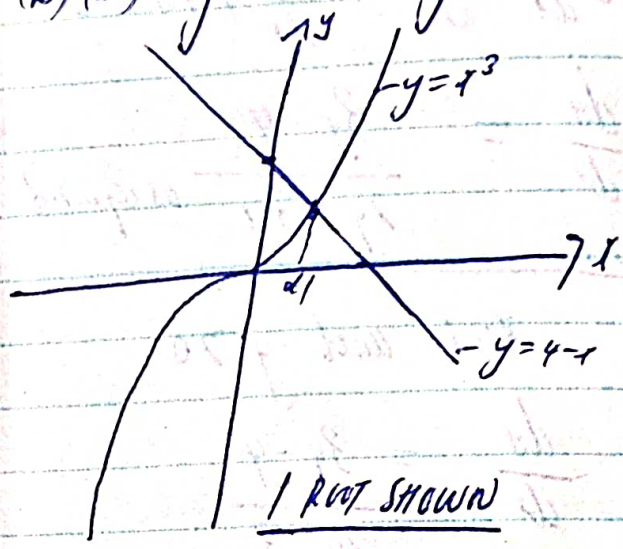
nearest degree

$PV = \pm 55°$

$\theta = 180n \pm 55°$

$\theta = [55°; 125°; 235°; 305°]$

(13)(ii) $y = x^3$ $y = 4 - x$



1 ROOT SHOWN

(ii) $f(x) = x^3 + x - 4$

$f(1) = (1)^3 + 1 - 4 = -2$

$f(1{,}5) = (1{,}5)^3 + (1{,}5) - 4 = 0{,}875$

∴ Since there's a sign change a root lies between 1 and 1,5

(b) $x_n = \sqrt{\frac{4}{x_n} - 1}$

$x_1 = 1{,}37$

$x_2 = \sqrt{\frac{4}{1{,}37} - 1}$

$x_2 = 1{,}3855$

$x_3 = 1{,}3736$

$x_4 = 1{,}3827$

$x_5 = 1{,}3758$

$\alpha = 1{,}38$

(ii) $y^2 - 3xy + 2x^2 = 6$

$$2y\frac{dy}{dx} - 3\left[(1)(y) + \frac{dy}{dx}(x)\right] + 4x = 0$$

$$2y\frac{dy}{dx} - 3y - 3x\frac{dy}{dx} + 4x = 0$$

$$\frac{dy}{dx} = \frac{3y - 4x}{2y - 3x}$$ as required

(b) $x = 1$ and $y > 0$

$$\frac{dy}{dx} = \frac{3y - 4}{2y - 3}$$

For x

$y^2 - 3y + 2 = 6$

$y^2 - 3y - 4 = 0$

$(y+1)(y-4) = 0$

$y = -1$ or 4

but $y > 0$

So we take $y = 4$

$$\frac{dy}{dx} = \frac{3(4) - 4}{2(4) - 3}$$

$$\frac{dy}{dx} = \frac{8}{5}$$

$P(1; 4)$

$$y - 4 = \frac{8}{5}(x - 1)$$

$$5y - 20 = 8x - 8$$

$$5y = 8x + 12$$

$$y = \frac{8}{5}x + \frac{12}{5}$$

$\Longrightarrow$

QED @ brainly